

class10

adding csv

```
pdb <- read.csv("pdb_stats.csv", row.names = 1)
pdb
```

	X.ray	EM	NMR	Integrative	Multiple.methods	Neutron
Protein (only)	178795	21825	12773	343	226	84
Protein/Oligosaccharide	10363	3564	34	8	11	1
Protein/NA	9106	6335	287	24	7	0
Nucleic acid (only)	3132	221	1566	3	15	3
Other	175	25	33	4	0	0
Oligosaccharide (only)	11	0	6	0	1	0
	Other	Total				
Protein (only)	32	214078				
Protein/Oligosaccharide	0	13981				
Protein/NA	0	15759				
Nucleic acid (only)	1	4941				
Other	0	237				
Oligosaccharide (only)	4	22				

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy

```
total_structures <- sum(pdb$Total)

xray_total <- sum(pdb$X.ray)
em_total <- sum(pdb$EM)

xray_pct <- (xray_total / total_structures) * 100
em_pct <- (em_total / total_structures) * 100

xray_pct
```

```
[1] 80.95077
```

```
em_pct
```

```
[1] 12.83843
```

structures in PDB using X-ray: 80.95077% and structures in PDB using EM: 12.83843%

Q2: What proportion of structures in the PDB are protein?

```
total_pdb <- sum(pdb$Total)
protein_total <- pdb["Protein (only)", "Total"]
```

```
protein_prop <- protein_total / total_pdb  
protein_prop
```

[1] 0.8596889

0.8596889 is the proportion of protein only which is 85.96%

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

Searching HIV in the website search box showed me that there are 1147 structures.

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

because x-ray crystallography does not resolve hydrogen good so only the oxygen atom is actually visible. And the oxygen atom is only 1 out of the 3 atoms molecules for water.

Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

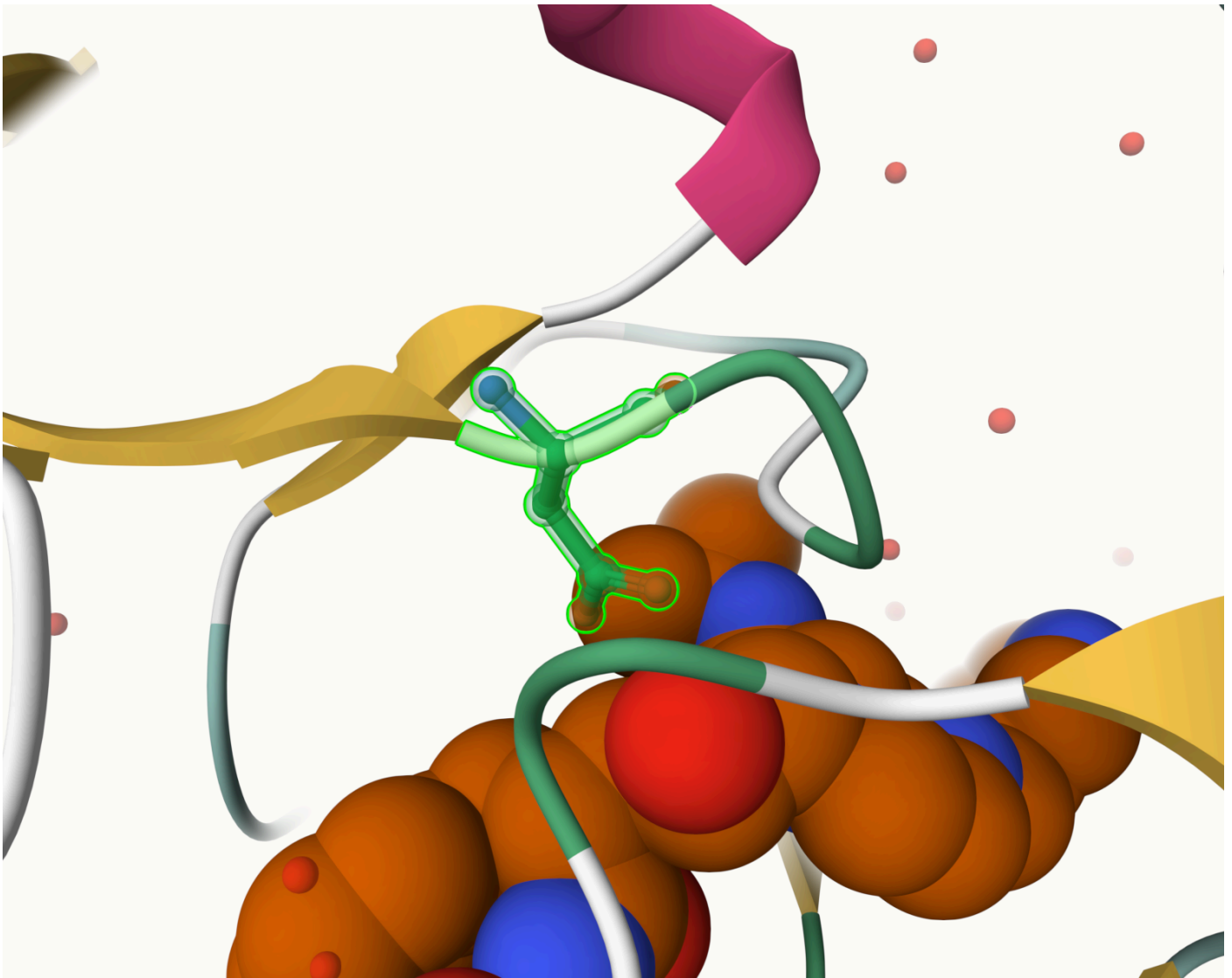
HOH 306

Q6. Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.

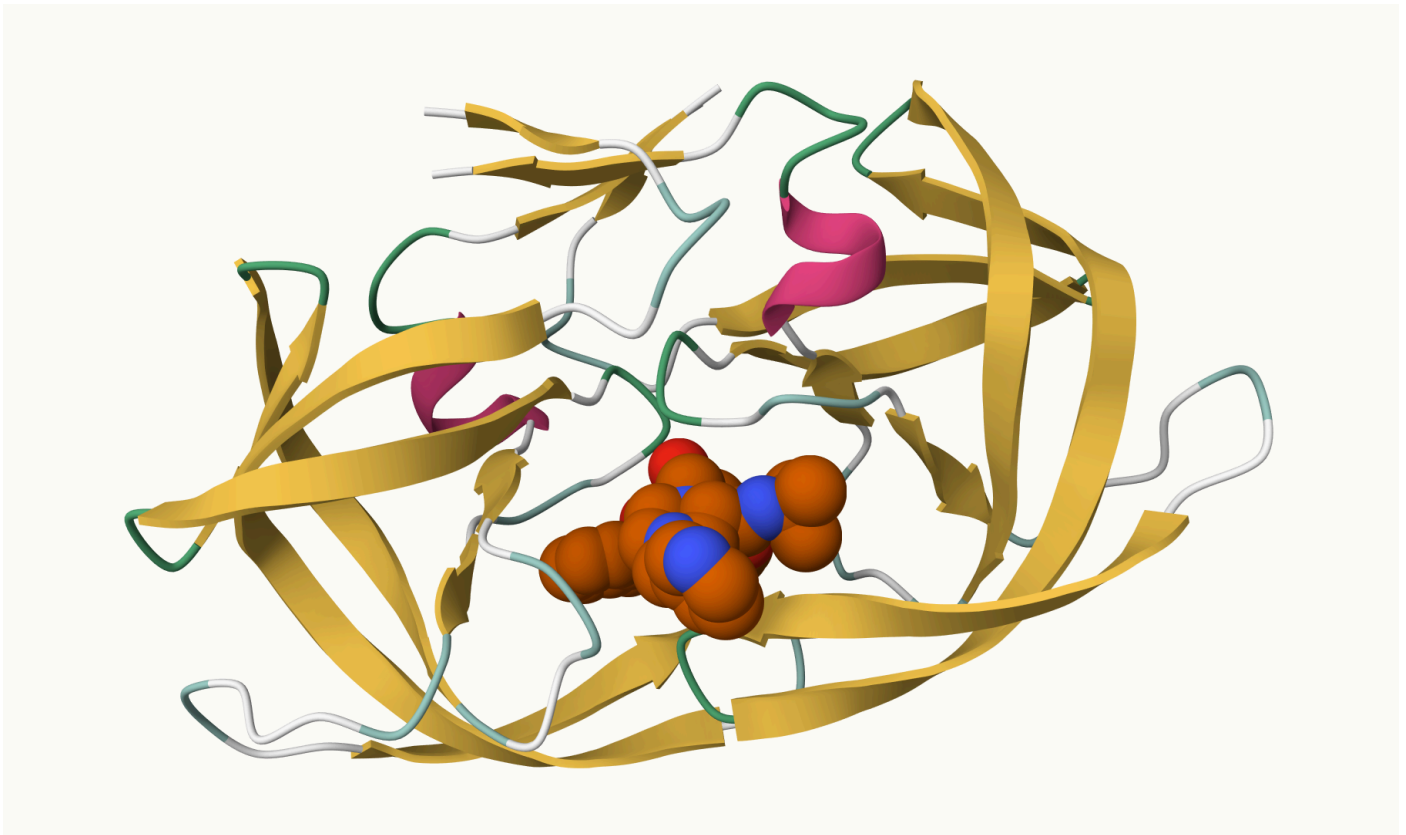
```
knitr::include_graphics("1HSG (1).png")
```



```
knitr::include_graphics("1HSG (2).png")
```



```
knitr::include_graphics("1HSG.png")
```



Q7: [Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

The four-stranded β -sheet at the dimer interface

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

pdb

Call: read.pdb(file = "1hsg")

```
Total Models#: 1
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

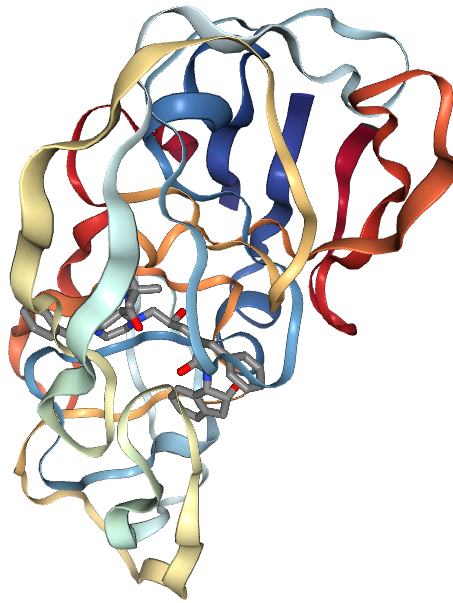
HOH

Q9: How many protein chains are in this structure?

2 chains

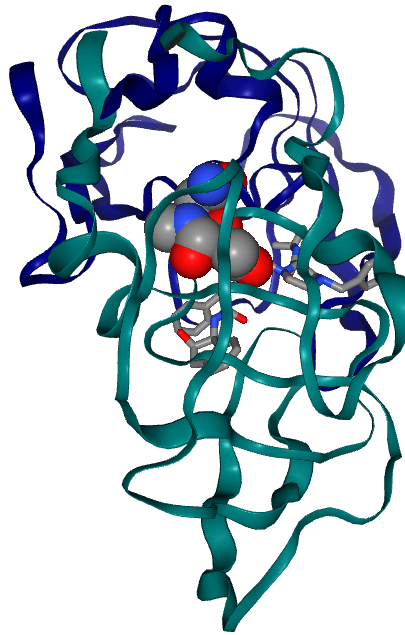
```
library(bio3dview)
library(NGLVieweR)

view.pdb(pdb) |>
  setSpin()
```



```
# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
view.pdb(pdb, cols=c("navy","teal"),
          highlight = sele,
          highlight.style = "spacefill") |>
setRock()
```



```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

Call: `read.pdb(file = "6s36")`

Total Models#: 1

Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)

Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

Protein sequence:

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPVKVEGDVDTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

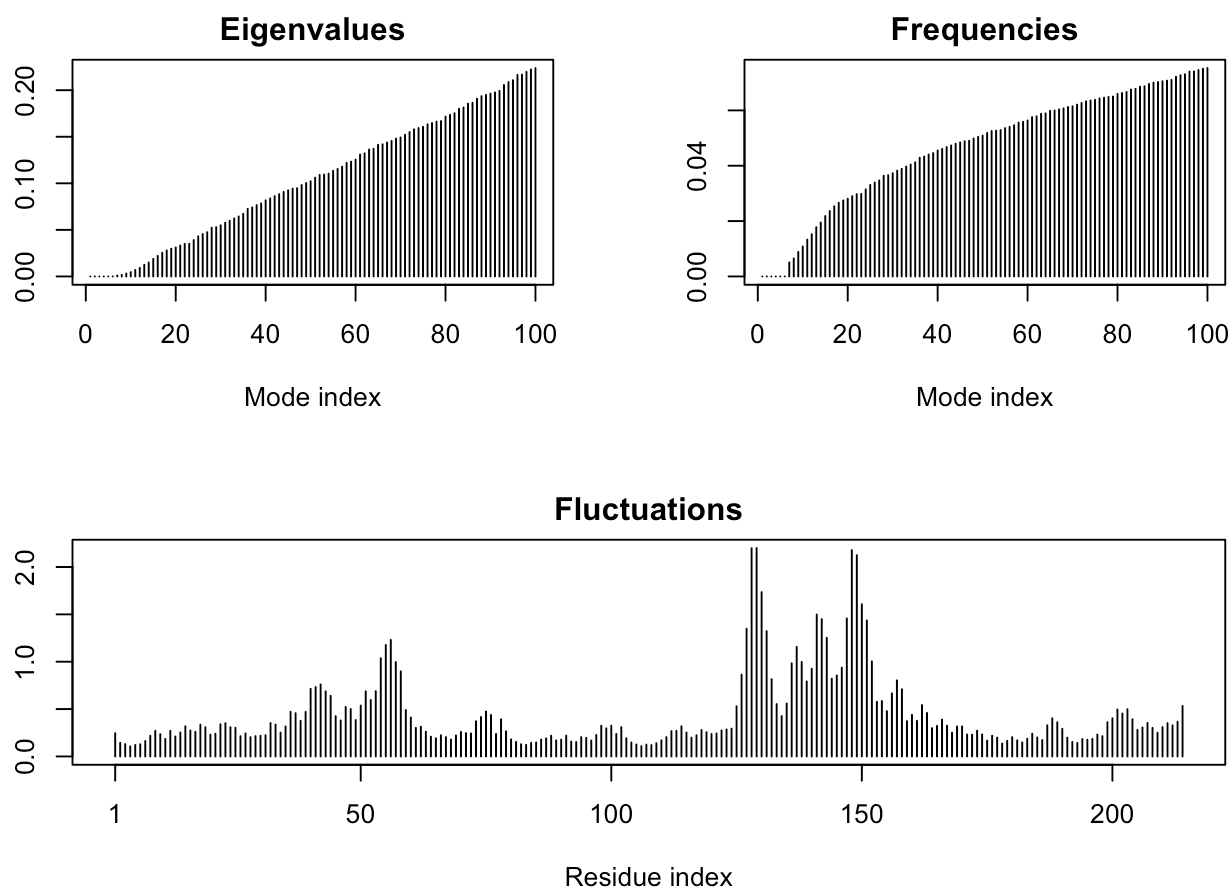


```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
# Perform flexibility prediction  
m <- nma(adk)
```

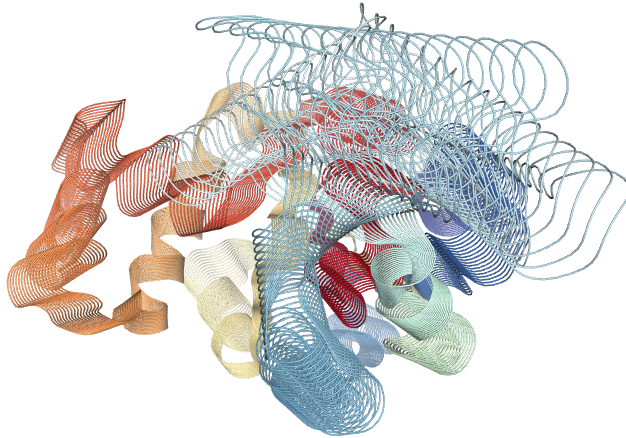
```
Building Hessian...      Done in 0.009 seconds.  
Diagonalizing Hessian... Done in 0.179 seconds.
```

```
plot(m)
```



```
mktrj(m, file="adk_m7.pdb")
```

```
view.nma(m, pdb=adk)
```



Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa

Q11. Which of the above packages is not found on BioConductor or CRAN?

bio3dview

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True

```
library(bio3d)
aa <- get.seq("lake_A")
```

Warning in get.seq("lake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214 amino acids

```
# Blast or hmmer search
#b <- blast.pdb(aa)
hits <- NULL
hits$ pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A', '1E4V_A', '5EJE_A', '1E4Y_A', '3X2S_A',

# Download related PDB files
files <- get.pdb(hits$ pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6S36.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/4PZL.pdb.gz exists. Skipping download

|
|

| 0%

=====		8%
=====		15%
=====		23%
=====		31%
=====		38%
=====		46%
=====		54%
=====		62%
=====		69%
=====		77%
=====		85%
=====		92%
=====		100%

```
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
```

```
PDB has ALT records, taking A only, rm.alt=TRUE
```

```
. PDB has ALT records, taking A only, rm.alt=TRUE
```

```
. PDB has ALT records, taking A only, rm.alt=TRUE
```

```
. PDB has ALT records, taking A only, rm.alt=TRUE
```

```
.. PDB has ALT records, taking A only, rm.alt=TRUE
```

```
.... PDB has ALT records, taking A only, rm.alt=TRUE
```

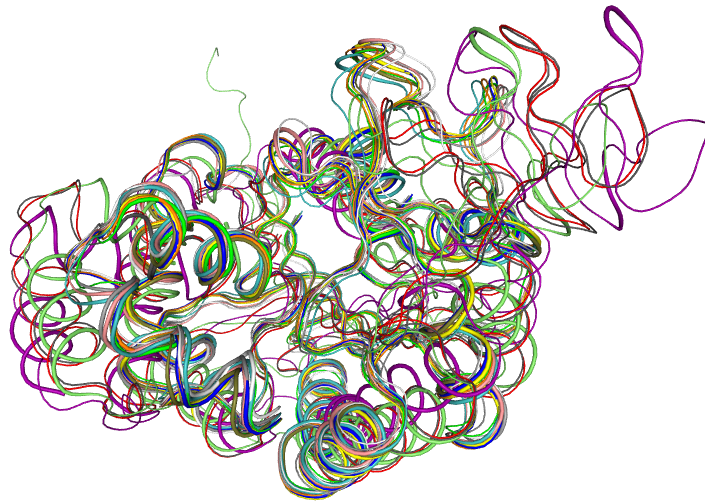
```
.    PDB has ALT records, taking A only, rm.alt=TRUE  
...
```

Extracting sequences

```
pdb/seq: 1    name: pdbs/split_chain/1AKE_A.pdb  
            PDB has ALT records, taking A only, rm.alt=TRUE  
pdb/seq: 2    name: pdbs/split_chain/6S36_A.pdb  
            PDB has ALT records, taking A only, rm.alt=TRUE  
pdb/seq: 3    name: pdbs/split_chain/6RZE_A.pdb  
            PDB has ALT records, taking A only, rm.alt=TRUE  
pdb/seq: 4    name: pdbs/split_chain/3HPR_A.pdb  
            PDB has ALT records, taking A only, rm.alt=TRUE  
pdb/seq: 5    name: pdbs/split_chain/1E4V_A.pdb  
pdb/seq: 6    name: pdbs/split_chain/5EJE_A.pdb  
            PDB has ALT records, taking A only, rm.alt=TRUE  
pdb/seq: 7    name: pdbs/split_chain/1E4Y_A.pdb  
pdb/seq: 8    name: pdbs/split_chain/3X2S_A.pdb  
pdb/seq: 9    name: pdbs/split_chain/6HAP_A.pdb  
pdb/seq: 10   name: pdbs/split_chain/6HAM_A.pdb  
            PDB has ALT records, taking A only, rm.alt=TRUE  
pdb/seq: 11   name: pdbs/split_chain/4K46_A.pdb  
            PDB has ALT records, taking A only, rm.alt=TRUE  
pdb/seq: 12   name: pdbs/split_chain/3GMT_A.pdb  
pdb/seq: 13   name: pdbs/split_chain/4PZL_A.pdb
```

```
library(bio3dview)
```

```
view.pdbs(pdbs)
```



```
# Vector containing PDB database codes
ids <- basename.pdb(pdb$ids)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli 0139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```
anno
```

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-ray
6S36_A	6S36	A	Protein	214	X-ray
6RZE_A	6RZE	A	Protein	214	X-ray
3HPR_A	3HPR	A	Protein	214	X-ray
1E4V_A	1E4V	A	Protein	214	X-ray
5EJE_A	5EJE	A	Protein	214	X-ray
1E4Y_A	1E4Y	A	Protein	214	X-ray

3X2S_A	3X2S	A	Protein	214	X-ray
6HAP_A	6HAP	A	Protein	214	X-ray
6HAM_A	6HAM	A	Protein	214	X-ray
4K46_A	4K46	A	Protein	214	X-ray
3GMT_A	3GMT	A	Protein	230	X-ray
4PZL_A	4PZL	A	Protein	242	X-ray

	resolution	scopDomain	pfam
--	------------	------------	------

1AKE_A	2.00	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
6S36_A	1.60	<NA>	Adenylate kinase, active site lid (ADK_lid)
6RZE_A	1.69	<NA>	Adenylate kinase, active site lid (ADK_lid)
3HPR_A	2.00	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4V_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
5EJE_A	1.90	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
3X2S_A	2.80	<NA>	<NA>
6HAP_A	2.70	<NA>	Adenylate kinase, active site lid (ADK_lid)
6HAM_A	2.55	<NA>	Adenylate kinase, active site lid (ADK_lid)
4K46_A	2.01	<NA>	Adenylate kinase, active site lid (ADK_lid)
3GMT_A	2.10	<NA>	<NA>
4PZL_A	2.10	<NA>	Adenylate kinase (ADK)

	ligandId
--	----------

1AKE_A	AP5
6S36_A	CL (3),NA,MG (2)
6RZE_A	NA (3),CL (2)
3HPR_A	AP5
1E4V_A	AP5
5EJE_A	AP5,C0
1E4Y_A	AP5
3X2S_A	JPY (2),AP5,MG
6HAP_A	AP5
6HAM_A	AP5
4K46_A	ADP,AMP,P04
3GMT_A	S04 (2)
4PZL_A	CA,FMT,GOL

	ligandName
--	------------

1AKE_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE
6S36_A	CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A	SODIUM ION (3),CHLORIDE ION (2)
3HPR_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE
1E4V_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE
5EJE_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE
3X2S_A	N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE
6HAM_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE
4K46_A	ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
3GMT_A	SULFATE ION (2)
4PZL_A	CALCIUM ION,FORMIC ACID,GLYCEROL

	source
--	--------

1AKE_A	Escherichia coli
6S36_A	Escherichia coli

6RZE_A Escherichia coli
 3HPR_A Escherichia coli K-12
 1E4V_A Escherichia coli
 5EJE_A Escherichia coli 0139:H28 str. E24377A
 1E4Y_A Escherichia coli
 3X2S_A Escherichia coli str. K-12 substr. MDS42
 6HAP_A Escherichia coli 0139:H28 str. E24377A
 6HAM_A Escherichia coli K-12
 4K46_A Photobacterium profundum
 3GMT_A Burkholderia pseudomallei 1710b
 4PZL_A Francisella tularensis subsp. tularensis SCHU S4

structureTitle

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBITOR AP5A REFINED AT 1.9 ANGSTROMS RESOLUTION: A MODEL FOR A CATALYTIC TRANSITION STATE

6S36_A

Crystal structure of E. coli Adenylate kinase R119K mutant

6RZE_A

Crystal structure of E. coli Adenylate kinase R119A mutant

3HPR_A

Crystal structure of V148G adenylate kinase from E. coli, in complex with Ap5A

1E4V_A

Mutant G10V of adenylate kinase from E. coli, modified in the Gly-loop

5EJE_A

Crystal structure of E. coli Adenylate kinase G56C/T163C double mutant in complex with Ap5a

1E4Y_A

Mutant P9L of adenylate kinase from E. coli, modified in the Gly-loop

3X2S_A

Crystal structure of pyrene-conjugated adenylate kinase

6HAP_A

Adenylate kinase

6HAM_A

Adenylate kinase

4K46_A

Crystal Structure of Adenylate Kinase from Photobacterium profundum

3GMT_A

Crystal structure of adenylate kinase from burkholderia pseudomallei

4PZL_A

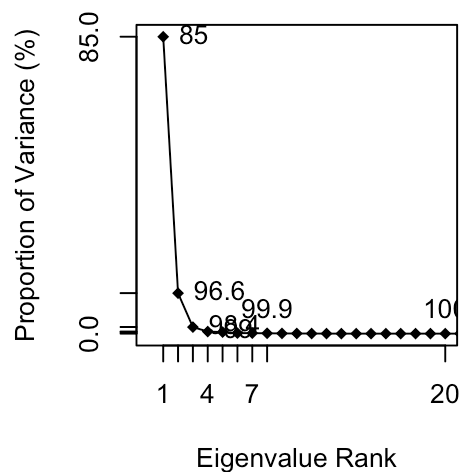
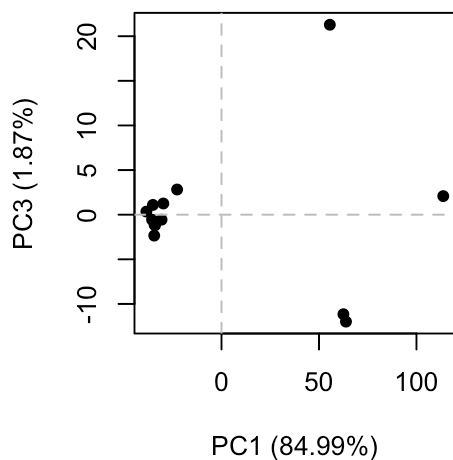
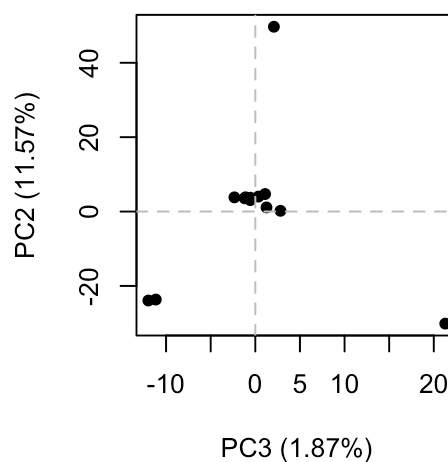
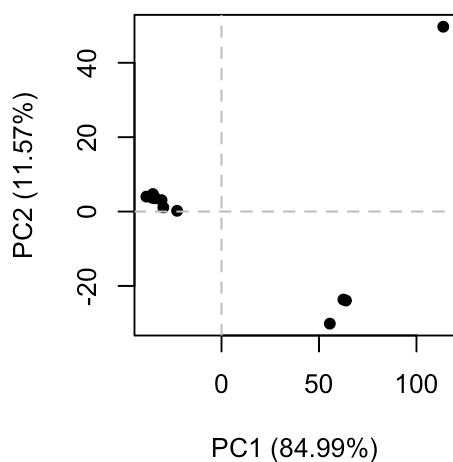
crystal structure of adenylate kinase from Francisella tularensis subsp. tularensis SCHU S4

		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al. J Mol Biology (1992)	0.19600	NA	
6S36_A	Rogne, P., et al. Biochemistry (2019)	0.16320	0.23560	
6RZE_A	Rogne, P., et al. Biochemistry (2019)	0.18650	0.23500	
3HPR_A	Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)	0.21000	0.24320	
1E4V_A	Muller, C.W., et al. Proteins (1993)	0.19600	NA	
5EJE_A	Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)	0.18890	0.23580	
1E4Y_A	Muller, C.W., et al. Proteins (1993)	0.17800	NA	
3X2S_A	Fujii, A., et al. Bioconjug Chem (2015)	0.20700	0.25600	

6HAP_A	Kantaev, R., et al. J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al. J Phys Chem B (2018)	0.20511	0.24325
4K46_A	Cho, Y.-J., et al. To be published	0.17000	0.22290
3GMT_A	Buchko, G.W., et al. Biochem Biophys Res Commun (2010)	0.23800	0.29500
4PZL_A	Tan, K., et al. To be published	0.19360	0.23680

	rWork	spaceGroup
1AKE_A	0.19600	P 21 2 21
6S36_A	0.15940	C 1 2 1
6RZE_A	0.18190	C 1 2 1
3HPR_A	0.20620	P 21 21 2
1E4V_A	0.19600	P 21 2 21
5EJE_A	0.18630	P 21 2 21
1E4Y_A	0.17800	P 1 21 1
3X2S_A	0.20700	P 21 21 21
6HAP_A	0.22370	I 2 2 2
6HAM_A	0.20311	P 43
4K46_A	0.16730	P 21 21 21
3GMT_A	0.23500	P 1 21 1
4PZL_A	0.19130	P 32

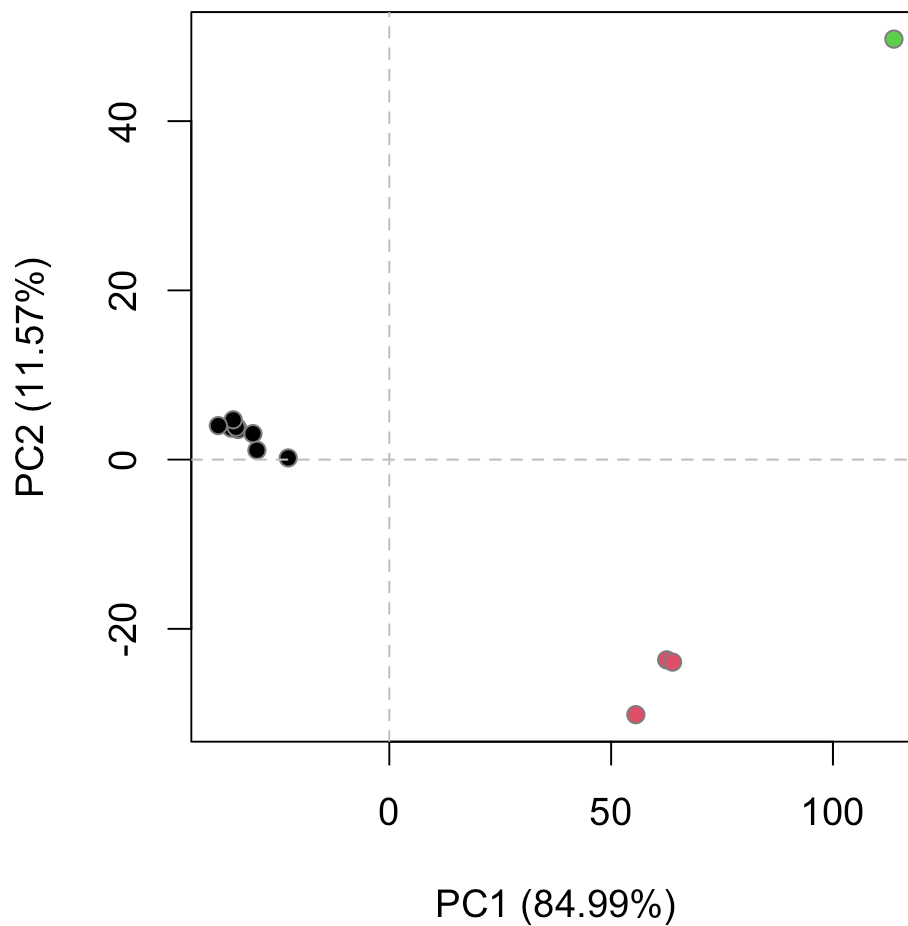
```
# Perform PCA
pc.xray <- pca(pdbx)
plot(pc.xray)
```



```
# Calculate RMSD  
rd <- rmsd(pdb)
```

Warning in rmsd(pdb): No indices provided, using the 204 non NA positions

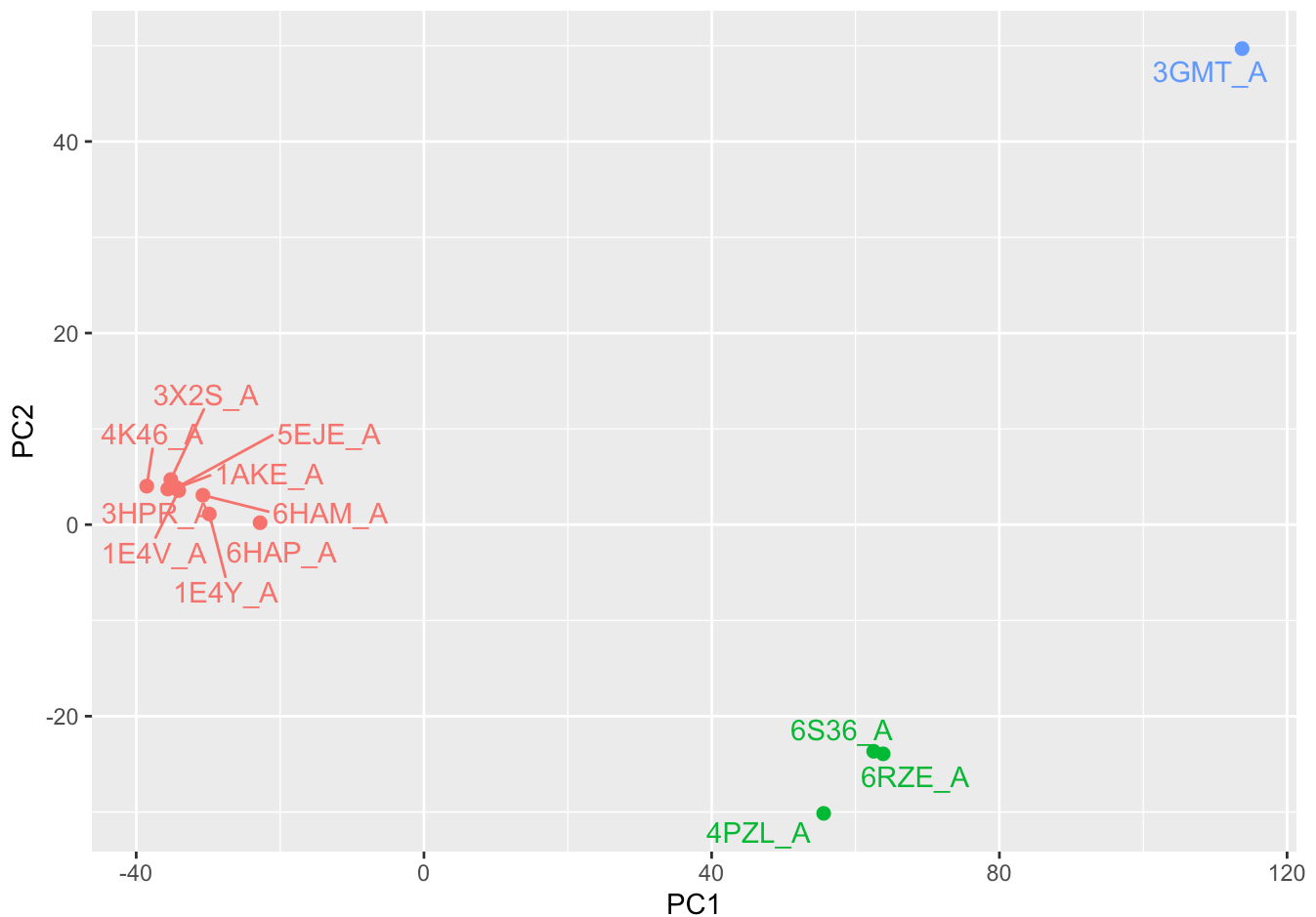
```
# Structure-based clustering  
hc.rd <- hclust(dist(rd))  
grps.rd <- cutree(hc.rd, k=3)  
  
plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



```
# Visualize first principal component  
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
#Plotting results with ggplot2  
library(ggplot2)  
library(ggrepel)  
  
df <- data.frame(PC1=pc.xray$z[,1],  
                 PC2=pc.xray$z[,2],  
                 col=as.factor(grps.rd),  
                 ids=ids)
```

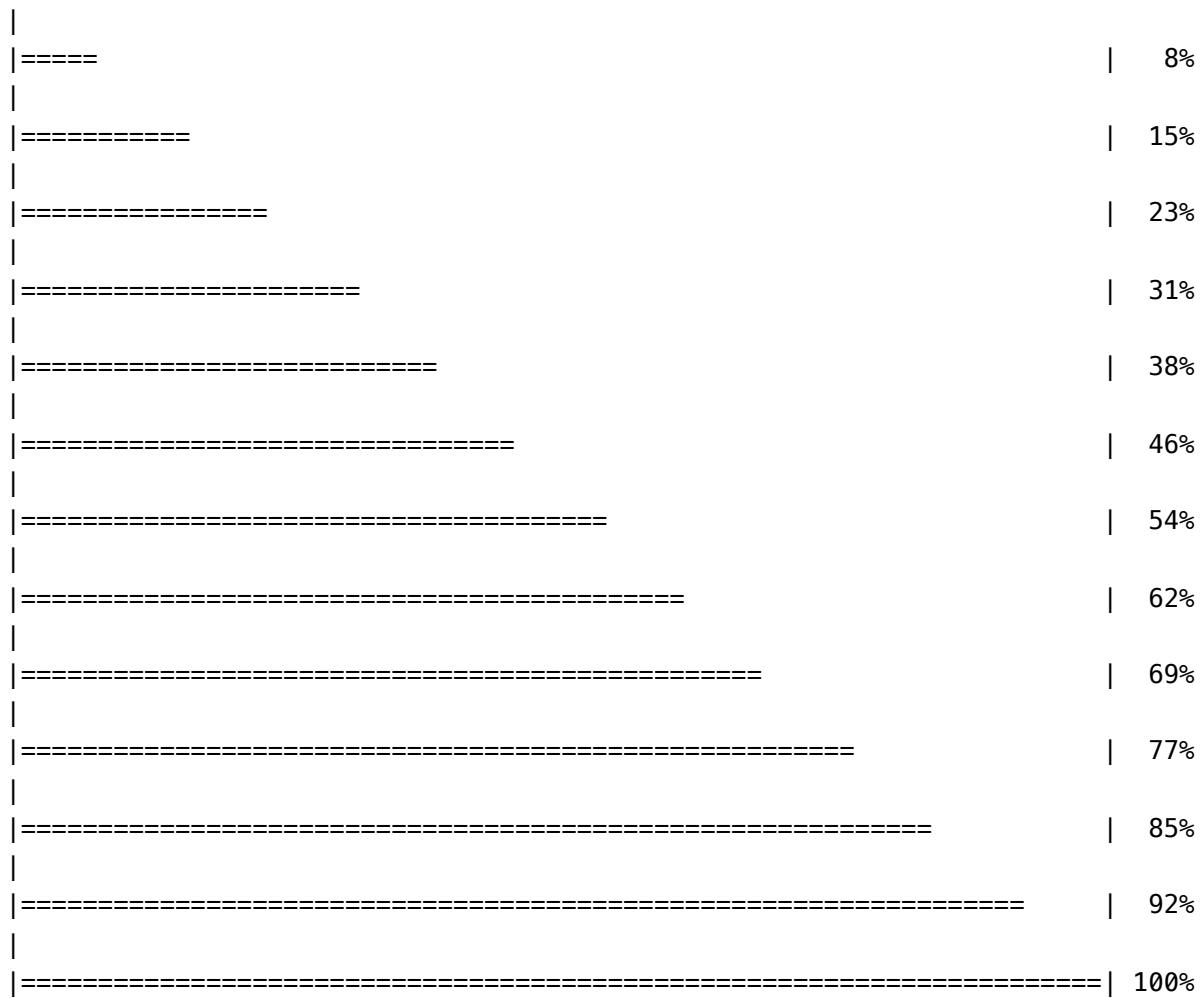
```
p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```



```
# NMA of all structures
modes <- nma(pdb)
```

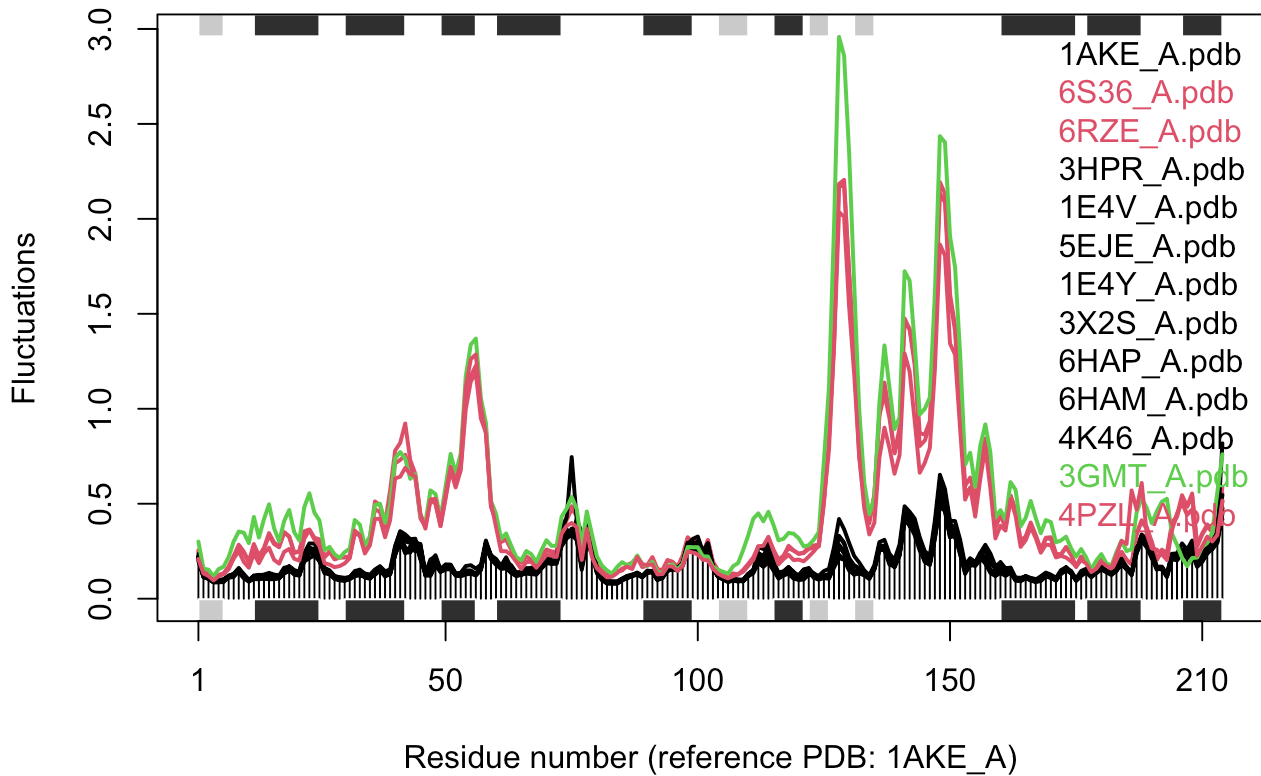
Details of Scheduled Calculation:

```
... 13 input structures
... storing 606 eigenvectors for each structure
... dimension of x$U.subspace: ( 612x606x13 )
... coordinate superposition prior to NM calculation
... aligned eigenvectors (gap containing positions removed)
... estimated memory usage of final 'eNMA' object: 36.9 Mb
```



```
plot(modes, pdba, col=grps.rd)
```

Extracting SSE from pdba\$sse attribute



Q14. What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

the black lines show lower fluctuation compared to the colored lines. the colored lines fluctuate around 30-50 and especially near 120-160 residues. These regions correspond to LID and AMPbinding domains, which are mobile domains that open and close. structures that do not have a nucleotide bound are more flexible in these regions.